

RODRIGO GAYTAN MIRANDA

Mechatronics Engineering | Industrial Automation, Electrical Systems & Industrial Operations

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SUMMARY

Mechatronics Engineer (Tecnológico de Monterrey, GPA 90/100) with hands-on experience in Hardware in the Loop (HIL) testing, Siemens TIA Portal PLC programming and industrial motor automation (NEMA/IEC). Currently validating embedded control systems and ECUs at SATVEN; previously at ABB México, collaborated across engineering and commercial teams to develop 10+ bilingual technical sales manuals, cutting response times by 10%. PMI-based project management specialization (Rice University) and international exposure from an academic exchange in Budapest.

WORK EXPERIENCE

HIL Engineer | SATVEN

Jun 2026 - Present

- Develop and execute Hardware-in-the-Loop (HIL) test suites for functional validation of electronic control units (ECUs) and embedded systems, designing automated test scenarios that verify system requirements with high coverage.
- Configure real-time simulation environments and signal conditioning hardware to accurately model vehicle dynamics and operating conditions. Perform root-cause analysis and fault isolation across hardware-software integration boundaries, logging and tracking defects through resolution.

Trainee specialist - Low and Medium Voltage Motors | ABB México

Jan 2025 - Jul 2025

- Developed 10+ bilingual technical sales manuals for NEMA, IEC, and Baldor motor product lines, reducing response times by 10% and supporting commercial teams.
- Prepared technical quotations for low and medium voltage motor solutions while supporting product selection processes and customer requirements.
- Collaborated across engineering and commercial teams to support product specification, technical documentation and industrial motor applications.

PROJECTS

Structures Senior - SAE Aerodesign | Tecnológico de Monterrey

Aug 2024 - Jun 2026

- Contributed to improving team performance from 22nd to 18th place among 33 teams at SAE Aerodesign East 2025 through structural design and prototype development.
- Designed and performed structural analysis of winglets to improve aerodynamic efficiency and flight performance of a remote controlled aircraft prototype.
- Manufactured and assembled aircraft structural components while supporting multidisciplinary design, integration, and testing activities.
- Remote/advisory role during academic exchange Aug 2025 - Jan 2026.

Flexible Single-link Robot Control System | CINVESTAV

Jan 2026 - Jun 2026

- Designed and programmed a robot featuring a Python-based reinforcement learning agent to autonomously tune optimal PID gains for flexible beam models.
- Achieved an 11% efficiency improvement in system response by suppressing structural vibrations and significantly reducing settling time.
- Engineered custom Python pipelines to process, analyze, and visualize real-time training metrics and control system performance.

PLC Programming- Scholarship Service | Tecnológico de Monterrey

Jan 2025 - June 2025

- Programmed and implemented industrial automation systems using Siemens S7 PLCs through TIA Portal, developing control logic with Function Block Diagram (FBD) programming.
- Integrated industrial communication networks using Profinet while designing automation systems involving pneumatic actuators, electrical actuators, HMIs and manufacturing cells.
- Developed automation and motion control applications including PID-controlled systems and motor control through variable frequency drives (VFDs), validating system integration and operational performance.

EDUCATION

B.S. in Mechatronics Engineering | Tecnológico de Monterrey - State of México Campus

Aug 2022 - Jun 2026

- Relevant coursework: Industrial automation, control systems, robotics, machine design, electronics.
- EGS Leadership Program: selective program focused on human leadership, entrepreneur spirit, and institutional values
- GPA : 90/100

International Exchange Program | Budapest University of Technology and Economics, Hungary

Aug 2025 - Jan 2026

- Completed coursework in Quality Management, electrical machines, renewable energy and manufacturing processes.

TECHNICAL SKILLS

- Automation & control:** PLC Siemens TIA Portal (FBD,LAD), Emulate3D
- Engineering Design:** SolidWorks (intermediate), CATIA (intermediate), AutoCAD (intermediate),Proteus 8
- Simulation & Analysis / Business tools:** Python (OpenCV, data analysis), C++ (basic), Git, GitHub, Power BI, Office 365, Simulink, MATLAB, Jira, Excel (advanced), Salesforce.
- Motor technologies:** NEMA,IEC low and medium voltage, technical specification and quotation
- Methodologies:** Industry 4.0, project management (PMI framework), Six sigma, quality management systems, lean manufacturing
- Languages:** Spanish (native), English (professional working proficiency)

CERTIFICATIONS

- Introduction to Cybersecurity | CISCO - 2025
- Engineering Project Management Specialization | Rice University (2026)